Quantifying the Rearrangement Complexity of Pangenomes

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Bielefeld University

May 24th, 2026

Rearrangements

INVERSIONS IN THE CHROMOSOMES OF DROSOPHILA PSEUDOOBSCURA*

TH. DOBZHANSKY AND A. H. STURTEVANT

California Institute of Technology, Pasadena, California

Received August 23, 1937

INDEPENDENT FUNCTIONS OF VIRAL PROTEIN AND NUCLEIC ACID IN GROWTH OF BACTERIOPHAGE*

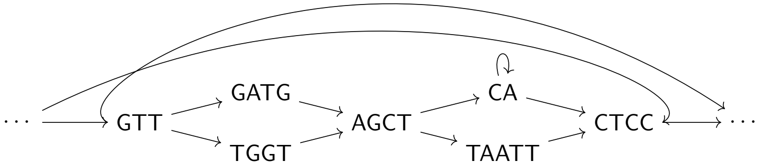
BY A. D. HERSHEY AND MARTHA CHASE

(*From the Department of Genetics, Carnegie Institution of Washington, Cold Spring Harbor, Long Island*)

(Received for publication, April 9, 1952)

- $\sim$30,000 inversions affect more bases of the human genome than $\sim$5,000,000 SNVs
- Pivotal role in:
  - Evolution
  - Disease
- Rearrangement Quantification for Structural Variants $\approx$ Alignment for SNVs

Abstracting from Local Variations



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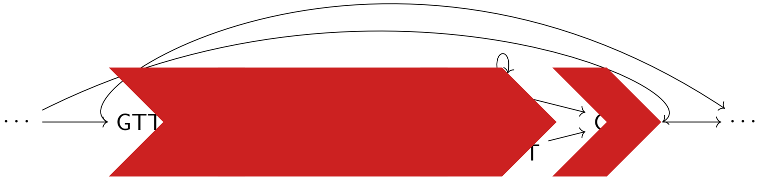

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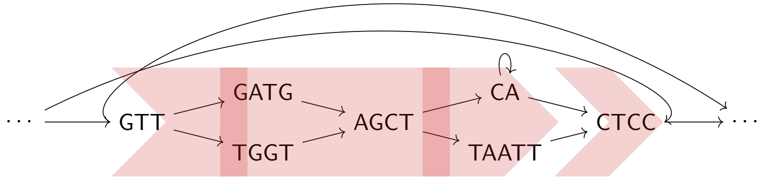

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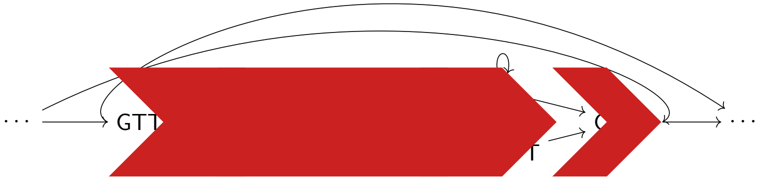

$\rightarrow$ Synteny Block Detection (more on Wednesday Session 4).

Marker, Chromosome, Genome, Pangenome

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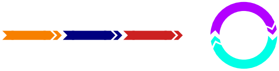

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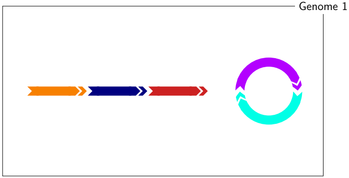

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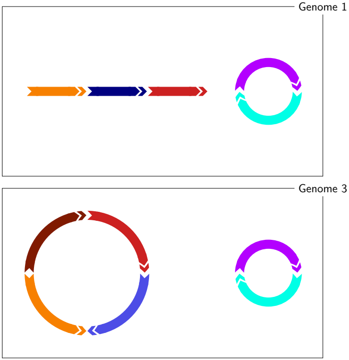

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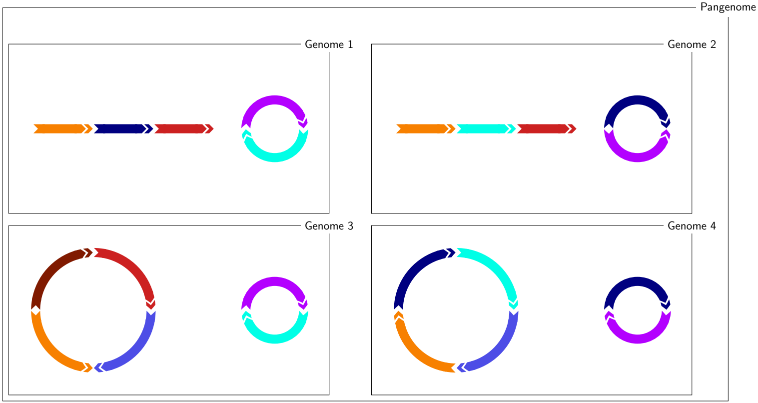

Multiple Breakpoint Graph (MBG)

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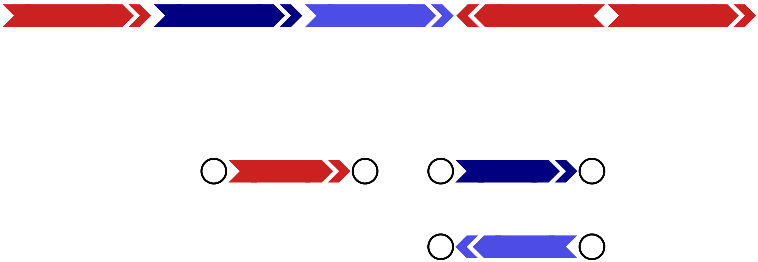

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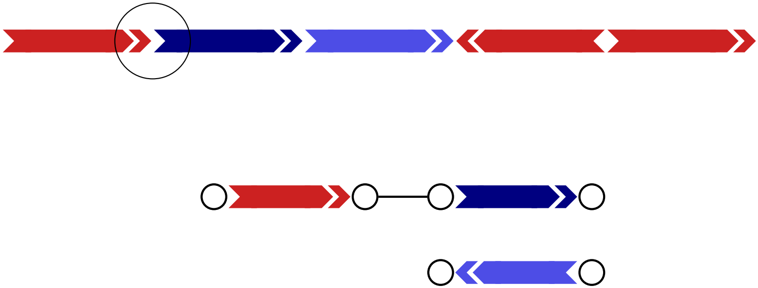

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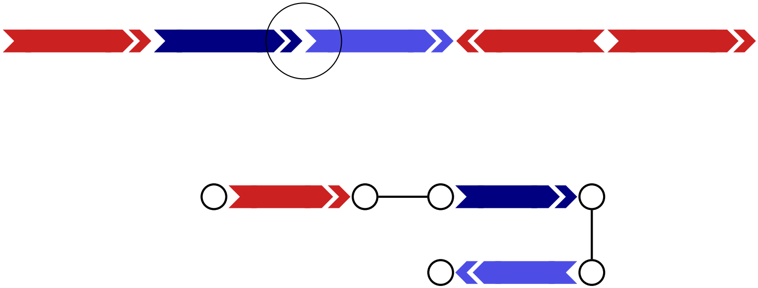

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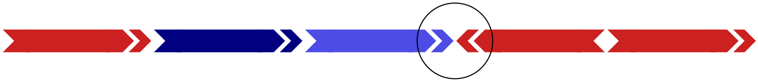

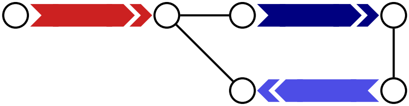

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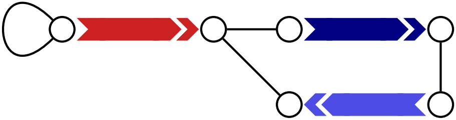

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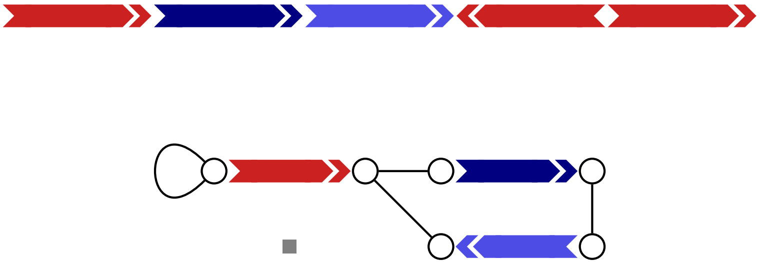

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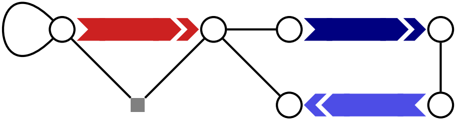
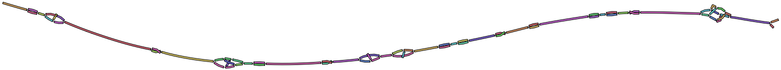


Quantifying Rearrangement Complexity in Pangenomes - Intuition



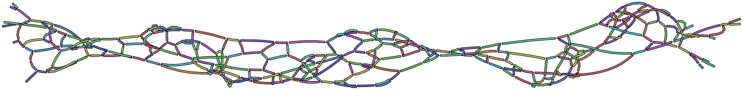


Image by courtesy of Luca Parmigiani

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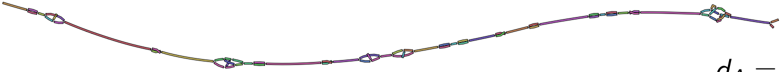
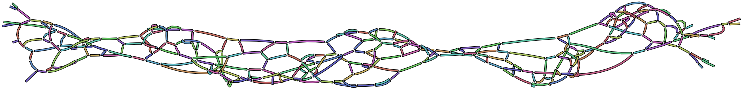


$d_A = 14$



$d_B = 107$

Image by courtesy of Luca Parmigiani

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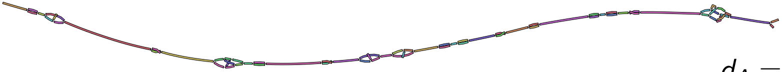
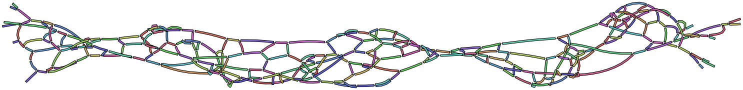


$d_A = ??$

$d_A < d_B$



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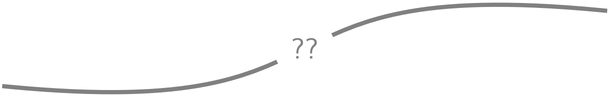
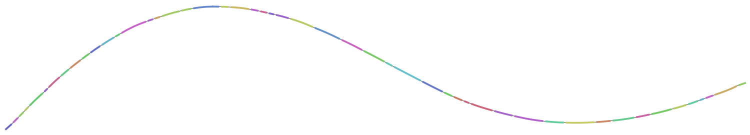
Quantifying Rearrangement Complexity in Pangenomes - Strategy



$d_X = 0$

Quantifying Rearrangement Complexity in Pangenomes - Strategy



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~~Boring~~ **Simple MBG**

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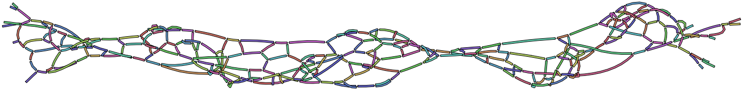

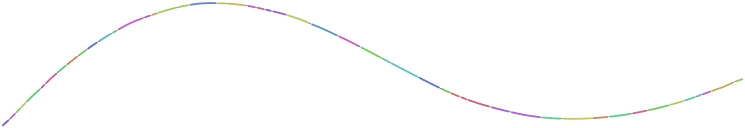

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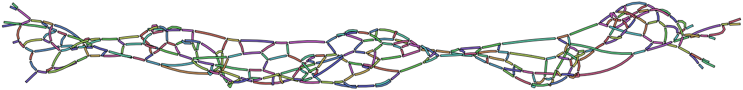
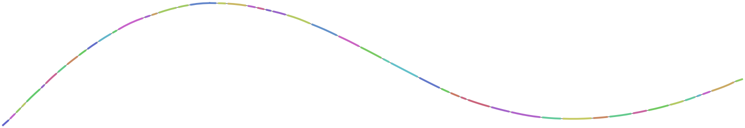


$d_B = |T|$

Transform by operations T



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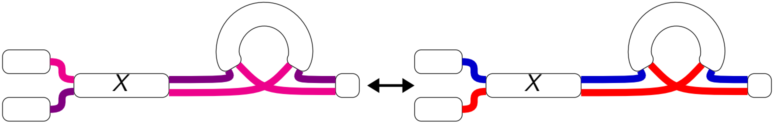
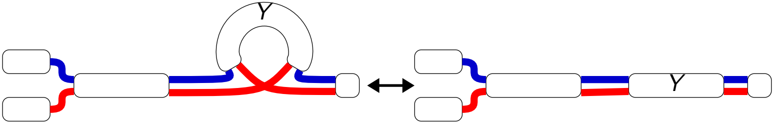
Path Altering Operations (PAOs)

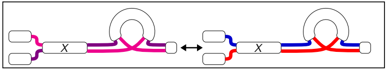


Graph Altering Operations (GAOs)

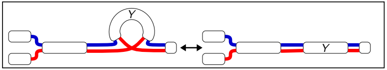


General Ancestral Reconstruction Problem (GARP)

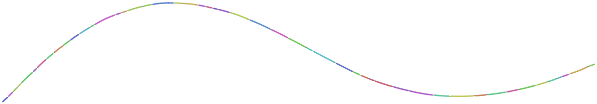
Use any number of PAOs



Minimize the number of GAOs



Simple pangenome



Rearrangement Distance Problem is a GARP-Specialization

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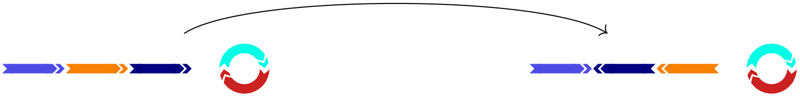

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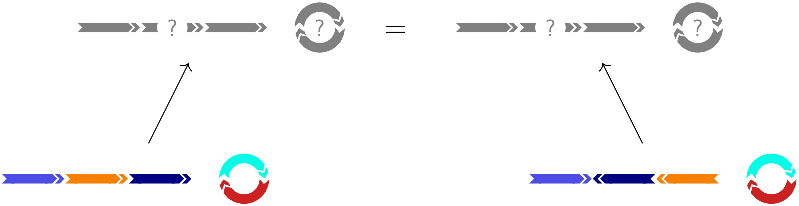

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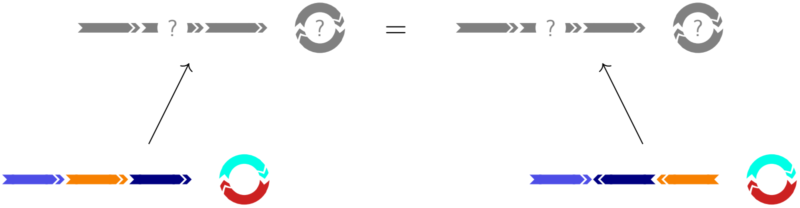
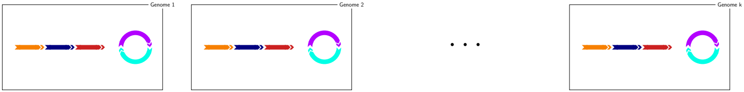


k-Coalescence – a Path Altering Operation



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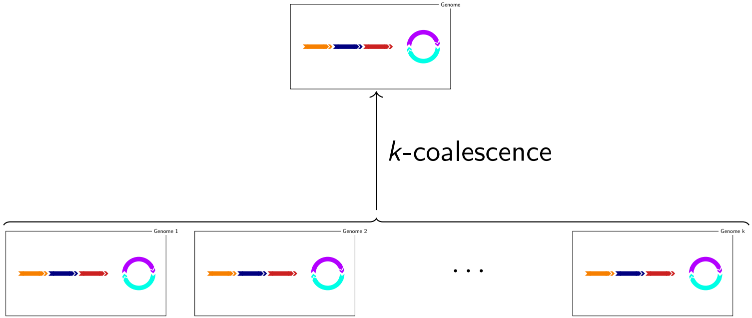
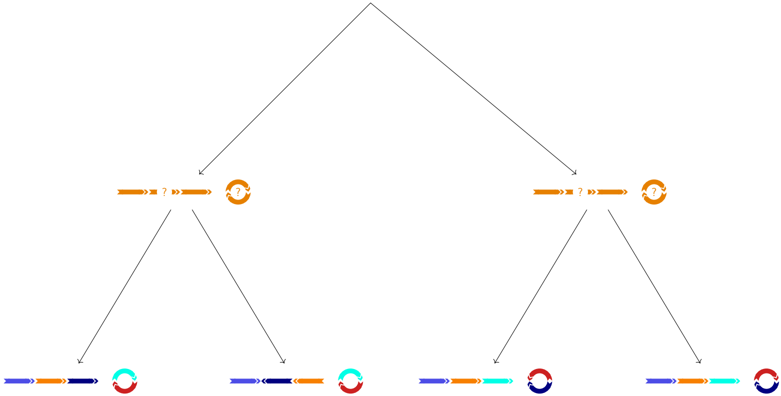


Large Parsimony Problem is a GARP Specialization



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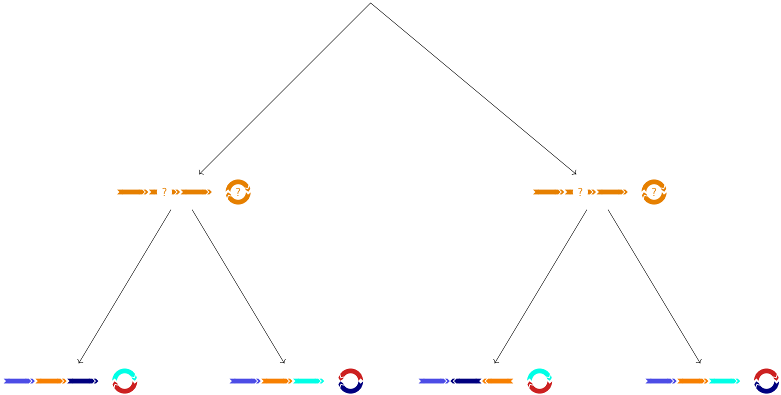

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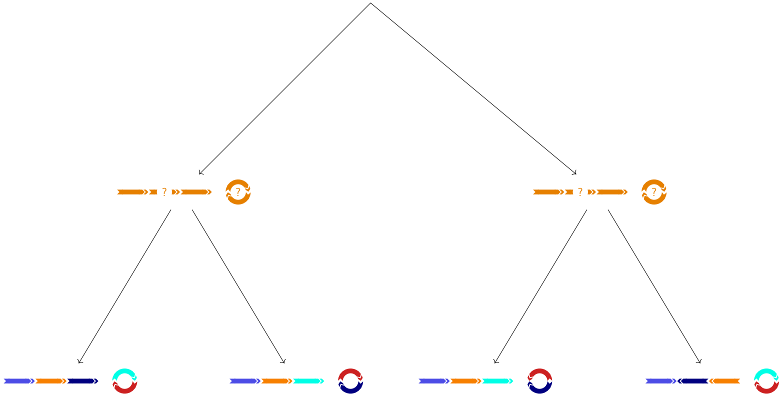

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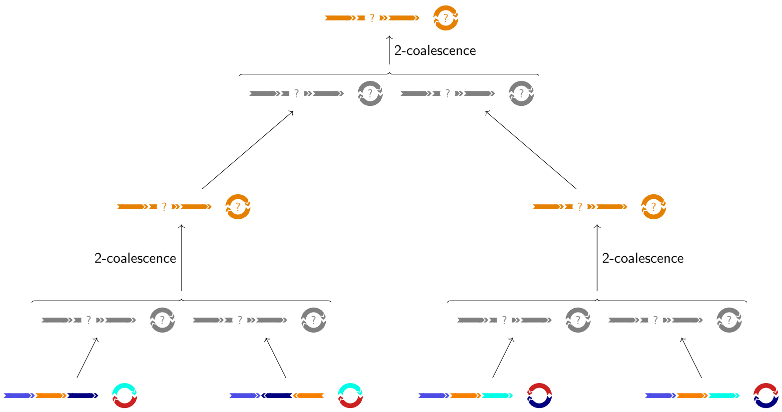


Relation to Classical Rearrangement Problems

Problem name	PAOs (N)
Distance problem	2-coalescence
Median problem	k-coalescence
Large parsimony problem	2-coalescence
Genome halving problem	genome halving
Guided genome halving problem	genome halving 2-coalescence

Classical Problem Formulations

- PAOs: mostly coalescence
- Very limited interaction between genomes
- Pangenomes $\rightarrow$ phylogenetic closeness $\rightarrow$ more interaction/horizontal effects

$\rightarrow$ Classical Problem Formulations would overestimate structural complexity for pangenomes (missing PAOs substituted with GAOs)

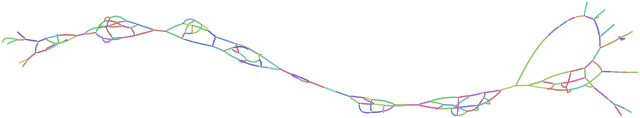
$\rightarrow$ For a lower bound, we need a "maximally powerful" set of PAOs

A maximally powerful set of PAOs

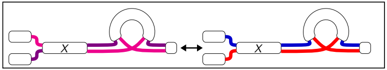
Definition

A set of PAOs N is called complete, if for all pairs of pangenomes $\mathbb{P}, \mathbb{P}'$ with the same MBG graph there is a sequence of PAOs $o_1 o_2 \ldots o_k \in N^*$, such that $\mathbb{P} \xrightarrow{o_1} \xrightarrow{o_2} \cdots \xrightarrow{o_k} \mathbb{P}'$.

Complete Ancestral Reconstruction for Pangenomes (CARP)



Use a complete set of PAOs



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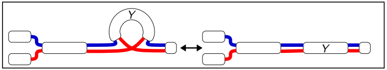
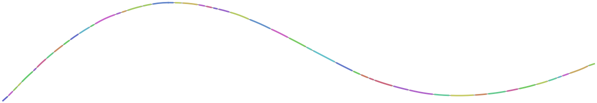
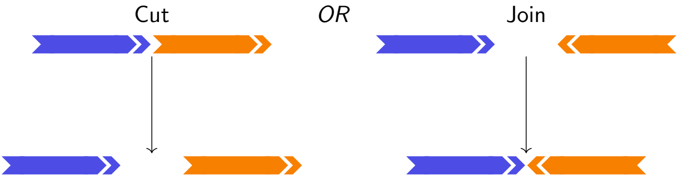


Simple pangenome



SCJ - A simple GAO



SCJ-CARP Theoretical result

- SCJ Large Parsimony is NP-hard

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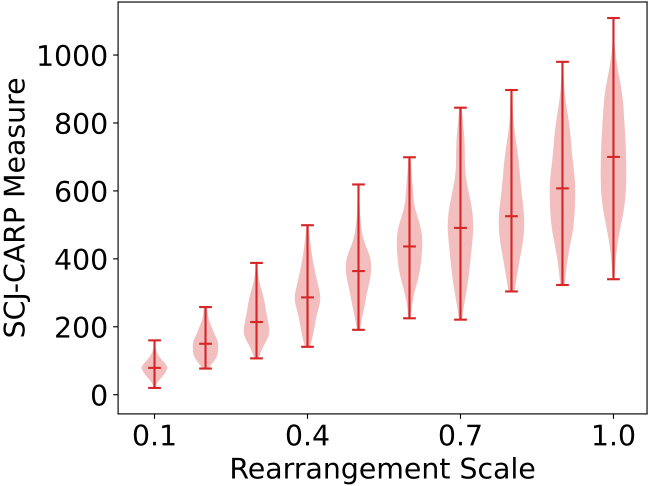
- SCJ Large Parsimony is NP-hard
- However SCJ-CARP:

Lemma

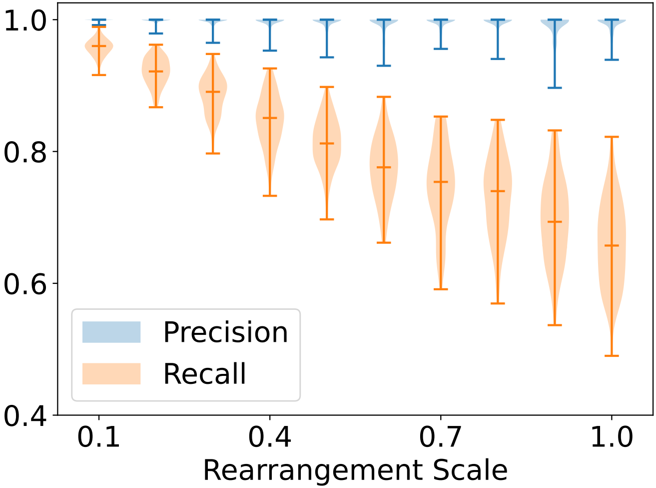
One needs exactly $|E_C|$ SCJs to transform a pangenome into a simple pangenome where $E_C \subset E_A$ is a subset of adjacency edges that can be determined in linear time.

SCJ-CARP Tracks Rearrangements

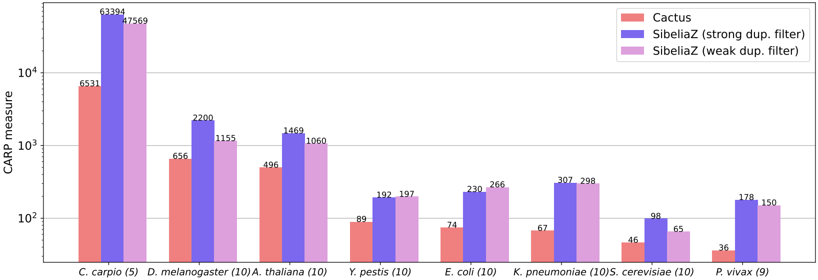


Pearson Coefficient: 0.88

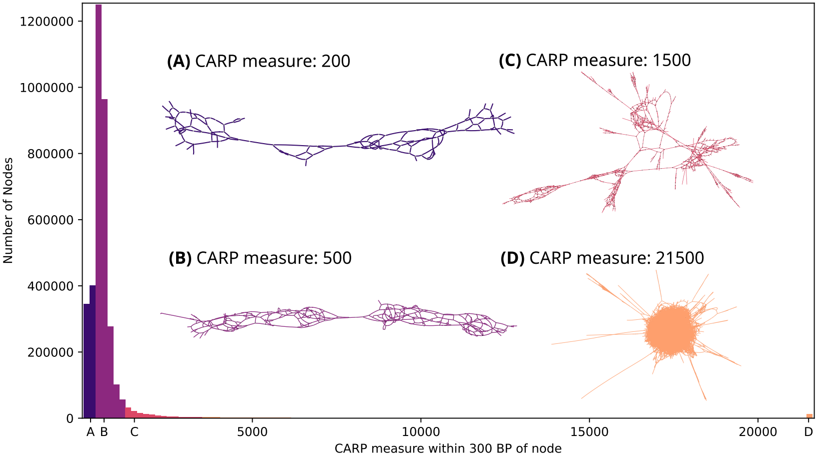
SCJ-CARP Reconstructs Ancestral Adjacencies (to a Limited Extent)



SCJ-CARP Pangenome Complexities



SCJ-CARP Region Complexities



CARP- Long Term Perspectives

- Tree/Lineages implied by CARP scenario

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  - $\rightarrow$ Characterize Pangenome by vector of CARP measures $(m_{SCJ}, m_{DCJ}, m_{HP}, m_{BI}, \ldots)$.
  - $\rightarrow$ Compare vectors to compare pangenomes

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  - $\rightarrow$ Compare vectors to compare pangenomes
- Comparing pangenomes $\mathbb{P}_a, \mathbb{P}_b$ on the same marker set
  - Among all possible CARP simplifications A_a, A_b
  - $d(\mathbb{P}_a, \mathbb{P}_b) = \min_{\mathbb{A}_a \in A_a, \mathbb{A}_b \in A_b} d(\mathbb{A}_a, \mathbb{A}_b)$

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- PAOs with a cost

Thank you!

slides

github